

Distance-to-Subspace Weights Disprove the Proposed Muckenhoupt–Reverse-Hölder Sufficiency

A full negative answer to the final question of arXiv:1910.13285

Automated proof attempt
agent_lane_03 model: GPT5.6

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Status: full counterexample, likely valid, pending human verification. The proposed sufficiency of $A_{p+\lambda/n} \cap RH_{n/(n-\lambda)}$ is false. In fact, a family of distance-to-subspace power weights in that class makes the Hardy–Littlewood maximal operator fail even from strong to weak weighted Morrey space.

Source question

For $1 \leq p < \infty$ and $0 < \lambda < n$, the source defines

$$\|f\|_{\mathcal{L}^{p,\lambda}(w)} = \sup_{x \in \mathbb{R}^n, r > 0} \left(r^{-\lambda} \int_{B(x,r)} |f|^p w \right)^{1/p},$$

and

$$\|g\|_{W\mathcal{L}^{p,\lambda}(w)} = \sup_{x,r,a>0} \left(r^{-\lambda} a^p w \{u \in B(x,r) : |g(u)| > a\} \right)^{1/p}.$$

After establishing shifted sharp ranges for point powers, the authors ask whether the whole class

$$A_{p+\lambda/n} \cap RH_{n/(n-\lambda)} \tag{1}$$

is sufficient in their Theorem 1.1 [1]. Since the pair $(|f|, Mf)$ satisfies the theorem’s weighted hypothesis, this would in particular imply Morrey boundedness of the Hardy–Littlewood maximal operator for every weight in (1). The appended crop reproduces the question exactly from page 16.

Proof intuition

The point powers treated in the source vanish at a zero-dimensional set. A ball larger than that small low-weight region pays for spreading its mass in every direction. The proposed weight classes do not encode this geometry.

Replace the point by a positive-dimensional subspace and use a power of the transverse distance. A thin tube is very cheap in weighted Morrey norm because the weight vanishes across its transverse thickness. Its ordinary Lebesgue mass is only one power of that thickness smaller, and the maximal operator spreads this mass over a fixed ball. If the subspace has enough tangential dimension to absorb the Morrey normalization, the two scalings are incompatible. The strict gap in the $A_{p+\lambda/n}$ range supplies exactly the exponents needed for the contradiction.

A weight-class calculation

We use the following standard subspace-power fact, including a short proof to fix all endpoint conventions.

Lemma 1 (Powers of transverse distance). *Write $\mathbb{R}^n = \mathbb{R}^d \times \mathbb{R}^m$, where $d + m = n$ and $m \geq 1$, and let $w(y, z) = |z|^\beta$. For $q > 1$,*

$$w \in A_q(\mathbb{R}^n) \iff -m < \beta < m(q - 1).$$

If $\beta \geq 0$, then $w \in RH_s(\mathbb{R}^n)$ for every finite $s > 1$.

Proof. For every $\gamma > -m$ and every ball $B = B((y_0, z_0), r) \subset \mathbb{R}^d \times \mathbb{R}^m$, scaling the transverse variable in the section integral gives

$$\frac{1}{|B|} \int_B |z|^\gamma dy dz \asymp (|z_0| + r)^\gamma. \quad (2)$$

Indeed, when $|z_0| \geq 2r$ the integrand is pointwise comparable to $|z_0|^\gamma$. When $|z_0| < 2r$, scale by r ; the remaining integrals have translation parameter in a compact set and are uniformly positive and finite precisely for $\gamma > -m$.

Apply (2) first with $\gamma = \beta$ and then with $\gamma = -\beta/(q - 1)$. Their powers cancel in the A_q product, and the second integral is locally finite exactly when $\beta < m(q - 1)$; local integrability of w itself requires $\beta > -m$. For RH_s , apply (2) with $\gamma = \beta s$ and $\gamma = \beta$. When $\beta \geq 0$ both are admissible and the two sides of the reverse Hölder inequality are comparable to $(|z_0| + r)^\beta$. $\square$

The counterexample family

Theorem 2 (Failure throughout a strict parameter strip). *Let $1 \leq p < \infty$, $0 < \lambda < n$, and write $\mathbb{R}^n = \mathbb{R}^d \times \mathbb{R}^m$ with*

$$d + m = n, \quad m \geq 1, \quad d \geq \lambda.$$

Choose

$$m(p - 1) < \beta < m \left(p - 1 + \frac{\lambda}{n} \right), \quad w(y, z) = |z|^\beta. \quad (3)$$

Then

$$w \in A_{p+\lambda/n} \cap RH_{n/(n-\lambda)},$$

but the Hardy–Littlewood maximal operator is not bounded

$$M : \mathcal{L}^{p,\lambda}(w) \longrightarrow W\mathcal{L}^{p,\lambda}(w).$$

Consequently it is not bounded on $\mathcal{L}^{p,\lambda}(w)$ either.

Proof. Put $q = p + \lambda/n$. The upper inequality in (3) is $\beta < m(q - 1)$, while the lower one gives $\beta > 0$. Lemma 1 therefore proves membership in both required classes.

For $0 < t < 1$, set

$$E_t = B_d(0, 1) \times B_m(0, t), \quad f_t = \chi_{E_t}.$$

We first estimate the strong input norm. Let $B = B((a, b), r)$ be any ball. If $r \leq t$, then on $B \cap E_t$ one has $w \leq t^\beta$, and hence

$$r^{-\lambda} \int_B f_t^p w \leq C r^{n-\lambda} t^\beta \leq C t^{n-\lambda+\beta} \leq C t^{m+\beta};$$

the last inequality uses $n - \lambda = d + m - \lambda \geq m$ and $t < 1$. If $t < r \leq 1$, transverse integration and the bound by a tangential d -ball of radius r give

$$r^{-\lambda} \int_B f_t^p w \leq C r^{d-\lambda} \int_{|z| < t} |z|^\beta dz \leq C t^{m+\beta},$$

because $d \geq \lambda$. Finally, if $r \geq 1$, use the total weighted mass of E_t and $r^{-\lambda} \leq 1$. Uniformly in all cases,

$$\|f_t\|_{\mathcal{L}^{p,\lambda}(w)} \leq C t^{(m+\beta)/p}. \quad (4)$$

For every $x \in B_n(0, 1)$, the ball $B_n(x, 3)$ contains E_t . Since $|E_t| = c_{d,m} t^m$, the centered (and therefore also the usual) Hardy–Littlewood maximal operator satisfies

$$M f_t(x) \geq c t^m \quad (x \in B_n(0, 1)). \quad (5)$$

Choose the weak-norm level $a = ct^m/2$ and the unit ball in its definition. Because $0 < w(B_n(0, 1)) < \infty$, (5) yields

$$\|Mf_t\|_{W\mathcal{L}^{p,\lambda}(w)} \geq c't^m. \quad (6)$$

Combining (4) and (6), the operator-norm quotient is at least

$$c''t^{m-(m+\beta)/p} = c''t^{[m(p-1)-\beta]/p} \longrightarrow \infty \quad (t \downarrow 0),$$

by the strict lower inequality in (3). This proves failure even into weak Morrey space. $\square$

Corollary 3 (A concrete strong-type disproof). *In $\mathbb{R}^2$, take*

$$p = 2, \quad \lambda = 1, \quad w(x_1, x_2) = |x_2|^{5/4}.$$

Then $w \in A_{5/2} \cap RH_2$, but M is not bounded on $\mathcal{L}^{2,1}(w)$; indeed it is not bounded from that space to $W\mathcal{L}^{2,1}(w)$.

Proof. Use Theorem 2 with $d = m = 1$ and $\beta = 5/4$. The two strict conditions are

$$1 = m(p-1) < \frac{5}{4} < \frac{3}{2} = m \left(p-1 + \frac{\lambda}{n} \right).$$

More explicitly, horizontal strips satisfy $|f_t|_{\mathcal{L}^{2,1}(w)} = O(t^{9/8})$, whereas $|Mf_t|_{W\mathcal{L}^{2,1}(w)} \geq ct$; their quotient grows as $t^{-1/8}$. $\square$

Scope, verification, and literature boundary

Corollary 3 alone fully answers the proposed universal sufficiency negatively, without relying on the weak endpoint $p = 1$. Theorem 2 shows that the obstruction is robust and even defeats the strong-to-weak estimate. The construction requires an integer codimension $m \leq n - \lambda$; it makes no claim about parameter regimes without such an m , nor about a replacement characterization of all admissible weights.

The script `code/check_scaling.py` checks the exact rational inequalities and the $t^{-1/8}$ quotient in the explicit example. It is not used in the proof. The ball estimates above were also checked at the three scale regimes $r \leq t$, $t < r \leq 1$, and $r \geq 1$; both boundary exponents match.

The bounded novelty audit searched the run indexes; exact class and source phrases; title/author citations; and hyperplane-power, subspace-distance, and weighted-Morrey maximal-operator terms through 11 August 2026. No matching resolution was found. The authors' later intrinsic-weight paper [2] gives necessary conditions and sufficiency with an extra local A_p hypothesis, while Lerner's 2022 note [3] still describes the broader global intrinsic characterization as open. Neither contains this construction. This is a bounded audit, not an exhaustive priority claim.

References

- [1] J. Duoandikoetxea and M. Rosenthal, *Boundedness of operators on certain power-weighted Morrey spaces beyond the Muckenhoupt weights*, Potential Anal. **53** (2020), 1255–1268; arXiv:1910.13285. Included as `source-paper.pdf`.
- [2] J. Duoandikoetxea and M. Rosenthal, *Muckenhoupt-type conditions on weighted Morrey spaces*, J. Fourier Anal. Appl. **27** (2021), Article 32; arXiv:2010.00250. Included as `supporting-2010.00250.pdf`.
- [3] A. K. Lerner, *A note on the maximal operator on weighted Morrey spaces*, arXiv:2211.07974 (2022). Included as `supporting-2211.07974.pdf`.

Exact source excerpt: final open question (page 16)

The range $\lambda - n < \beta < \lambda + n(p - 1)$ for weights of type $|x|^\beta$ corresponds to all the power weights in $A_{p+\lambda/n} \cap RH_{n/(n-\lambda)}$. On the other hand, the result in Theorem 1.1 is valid for $\lambda < n/\sigma'$, that is, we need $w \in RH_\sigma$ for some $\sigma > n/(n - \lambda)$ to get the estimate. Such a σ exists for any weight in $RH_{n/(n-\lambda)}$ by the self-improvement property of the reverse Hölder inequalities (Gehring's lemma). The endpoint weight $|x|^{\lambda-n}$, for which the estimates for M hold, is not in $RH_{n/(n-\lambda)}$, but it is in RH_σ for every $\sigma < n/(n - \lambda)$. One could guess the necessity of a reverse Hölder condition of this type in Proposition 5.2, but we have not been able to get it.

The weights in $A_{p+\lambda/n} \cap RH_{n/(n-\lambda)}$ are characterized by the factorization $u^{-\lambda/n}w$ with $u \in A_1$ and $w \in A_p \cap RH_{n/(n-\lambda)}$. The result in Theorem 1.1 covers all the weights of this type for which u is a power weight in A_1 . By translation invariance power weights can be taken to be centered at a point different from the origin. The sufficiency of $A_{p+\lambda/n} \cap RH_{n/(n-\lambda)}$ in Theorem 1.1 remains an open question for us.